Recovering the Source of AI-Transformed Text: Retrieval and Residual Traces in a Digital Collection

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Abstract

Digital libraries may encounter AI-transformed texts without reliable links to the collection records that guided their generation. We study *post-hoc source-link recovery*: using a generated derivative to rank its conditioning source within a known collection. Our controlled benchmark contains 60 passages generated from three King James Version chapters by two chat systems under content-preserving and style-oriented instructions. Each passage queries all 260 New Testament chapters using transparent lexical, character-based, stylometric, and hybrid retrieval methods. Word TF–IDF achieves mean reciprocal rank (MRR) of .967 for content adaptations and .956 for style-oriented transformations. After masking words participating in source-shared 3-grams, MRR remains .944 for content passages but falls to .598 for style-oriented passages. Stylometric MRR for style-oriented passages falls from .540 to .018, indicating substantial dependence on retained source-local phrasing. These results provide exploratory evidence that lightweight retrieval can support source-link suggestions in a controlled collection. The small, homogeneous dataset does not establish transferability to larger or more diverse digital libraries.

CCS Concepts

• **Information systems** → **Digital libraries and archives**; **Information retrieval**.

Keywords

digital libraries, source attribution, provenance, information retrieval, generative AI, stylometry, text reuse

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1 Introduction

Generative AI systems can transform digital-library materials through summarization, modernization, paraphrasing, and stylistic imitation. Although these transformations may support accessibility and creative reuse, the resulting texts can circulate without reliable metadata linking them to the collection records that informed their generation. Recovering such links could support provenance display, curator review, text-reuse discovery, and navigation between generated derivatives and canonical materials.

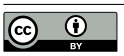
Emerging transparency mechanisms primarily address provenance at generation time. Article 50 of the European Union Artificial Intelligence Act requires providers of systems generating synthetic text and other media to support machine-readable marking and detection of artificially generated or manipulated outputs [3]. Language-model watermarking similarly embeds detectable signals during generation [4], while the C2PA specification provides infrastructure for recording authenticated content provenance and modification history [2]. These mechanisms are valuable, but markings or metadata may be absent, removed, or unavailable to a collecting institution. They also identify that content was generated or modified without necessarily identifying the particular collection record that conditioned a derivative.

We study this residual case as *post-hoc source-link recovery*: given an AI-transformed passage and a known collection containing its conditioning source, rank the collection records using only the derivative text. A content-preserving transformation may retain the source's ideas and rhetorical sequence while replacing its wording. A style-oriented transformation may replace the topic while attempting to preserve sentence patterns, repetition, pacing, and discourse organization. The latter setting is especially challenging because semantic continuity between the derivative and source is intentionally reduced.

Related research has treated missing-source attribution and historical text reuse as retrieval problems [5, 10]. Work on grounded generation has instead connected claims to supporting documents or produced answers with explicit citations [6, 9]. Much of this work either assumes semantic continuity or creates attribution during generation. Post-hoc source-link recovery begins after the derivative has already been produced and explicit generation evidence is unavailable.

This poster presents an exploratory, closed-collection study using all 260 chapters of the King James Version New Testament. Two chat systems produced 60 passages from three selected source chapters under content-preserving and style-oriented instructions. Each passage is used to rank all candidate chapters with transparent lexical, character-based, stylometric, and hybrid representations. Rather than proposing a state-of-the-art retrieval model, the study asks whether residual textual evidence is sufficient for source recovery in a controlled collection and which signals remain after directly shared phrases are removed.

Word TF–IDF achieves MRR of .967 for content adaptations and .956 for style-oriented transformations before masking. After words participating in source-shared 3-grams are replaced, MRR remains



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.944 for content passages but falls to .598 for style-oriented passages. Stylometric MRR for style-oriented passages decreases from .540 to .018. These results provide an initial proof of concept and show that apparently stylistic recovery can depend heavily on retained source-local phrasing. They do not establish transferability to larger, heterogeneous, multilingual, or open-world collections.

2 Dataset and Experimental Design

2.1 Collection and Source Selection

The retrieval collection consists of all 260 chapters of the King James Version New Testament, with each chapter treated as a separate candidate record. Markup and verse numbering were removed while punctuation and paragraph structure were retained.

This collection provides clearly defined record boundaries and allows the conditioning source of every generated passage to be established unambiguously. However, it is deliberately narrow and homogeneous: all candidate records come from the same translation and share a broadly similar historical register, vocabulary, and religious domain. The experiment should therefore be interpreted as a controlled test of within-collection source discrimination, rather than evidence of performance in heterogeneous or open-world digital libraries.

Three chapters were selected as conditioning sources: John 1, Hebrews 1, and Luke 15. They represent different forms of discourse, including theological exposition, doctrinal argument, and narrative. The source set nevertheless remains small, and the study does not test multilingual collections, multi-document derivation, or transformations based on sources outside the retrieval collection.

2.2 Generation Protocol

Two proprietary chat systems were accessed through their user interfaces on July 15, 2026. Because their exact backend configurations were not exposed, we report the displayed labels, *ChatGPT 5.6* and *Gemini 3.5 Thinking*, without treating them as stable model identifiers. Each prompt included one complete source chapter and requested five separately labeled passages.

The *content-preserving* condition requested a contemporary-English adaptation that retained the source's central ideas, events or claims, and rhetorical sequence while avoiding direct quotation. The *style-oriented* condition replaced the original topic with a coastal community preparing for a severe storm, while asking the model to preserve features such as repetition, sentence structure, pacing, and discourse organization. For example, when conditioned on John 1, the content task retained its theological subject matter, whereas the style task discussed an unrelated storm using a similar rhetorical progression.

The design contains two systems, three source chapters, and two conditions. The 12 prompts each returned five passages, yielding 60 passages. First responses were retained without regeneration or manual rewriting. Because the five passages within each response may be correlated, we treat them as 12 generation batches rather than 60 independent model calls.

3 Retrieval and Evaluation

3.1 Representations

Each generated passage queries the full 260-chapter collection. We compare five unsupervised representations. **BM25** uses $k_1 = 1.2$ and $b = .75$ [7]. **Word TF–IDF** uses lowercased word unigrams and bigrams, sublinear term frequency, and cosine similarity [8]. **Character TF–IDF** uses character 3–5-grams, a 50,000-feature cap, and cosine similarity.

Stylometry contains 152 function- and frequent-word rates plus 13 features for sentence length, word length, lexical diversity, paragraphs, and punctuation. Features are standardized on the candidate chapters and ranked by Euclidean distance, following the content-light logic of conventional stylometry [1, 11]. **Hybrid** averages per-query standardized Word TF–IDF and stylometric similarities. We additionally evaluate the 13 non-word features alone as a *structural profile*; this post-hoc diagnostic is not included in the hybrid.

3.2 Phrase-Masking Ablation

To test whether successful recovery depends on directly retained phrases, we conduct an oracle ablation using the known correct source. For each query and $n \in \{3, 4, 5\}$, every query word belonging to an exact source-shared n-gram is replaced with an artificial alphabetic token of the same length. The replacements are assigned mechanically rather than selected for semantic similarity, and repeated occurrences of the same original word receive the same replacement. This removes the lexical identity of shared phrases while preserving word length, punctuation, sentence boundaries, and approximate repetition patterns.

Because the procedure requires access to the correct source, it is used only to diagnose which textual signals support retrieval and is not proposed as a deployable retrieval method. Under the strict 3-gram condition, masking affects an average of 7.0% of content tokens and 36.4% of style-oriented tokens. For style-oriented passages, the masked proportion is 14.8% for ChatGPT and 57.9% for Gemini, indicating substantial variation between the two systems.

3.3 Metrics and Inference

We report MRR, Recall@1, Recall@5, and median rank. With 260 candidates, random ranking has expected MRR .024 and Recall@1 .004. Passage-level metrics are descriptive. For condition comparisons, reciprocal ranks are first averaged within each batch, producing six matched content/style pairs defined by two systems and three sources. We enumerate all $2^6 = 64$ within-pair label swaps for an exact two-sided test. Masking comparisons use the same batch-level paired procedure.

4 Results

4.1 Baseline Recovery and Transformation Type

Table 1 answers RQ1. Word TF–IDF retrieves the correct chapter at rank one for 28 of 30 passages in each condition, reaching MRR .967 for content and .956 for style. The hybrid places every target in the top five. BM25 is less stable under style transformation (MRR .666), whereas character TF–IDF remains similar across conditions.

Table 1: Source-chapter retrieval among 260 candidates. Each condition contains 30 passages in six generation batches.

	Content			Style		
Method	MRR	R@1	R@5	MRR	R@1	R@5
BM25	.902	.867	.933	.666	.633	.667
Word TF–IDF	.967	.933	1.00	.956	.933	.967
Character TF–IDF	.865	.833	.900	.880	.867	.867
Stylometry	.072	.033	.133	.540	.467	.567
Hybrid	.942	.900	1.00	.967	.933	1.00

Table 2: MRR before and after masking source-shared phrases.

Representation	Condition	Orig.	M5	M4	M3
Word TF–IDF	Content	.967	.967	.967	.944
Word TF–IDF	Style	.956	.938	.913	.598
Stylometry	Content	.072	.070	.050	.042
Stylometry	Style	.540	.049	.036	.018
Structural	Style	.353	.239	.245	.267

M5, M4, and M3 mask words participating in source-shared 5-, 4-, and 3-grams, respectively.

These results show that inexpensive sparse retrieval can generate a short source-candidate list in this controlled collection.

For RQ2, stylometry produces the largest contrast. Its MRR rises from .072 for content passages to .540 for style passages, and median rank improves from 103 to 2. All six model–source pairs have higher batch MRR in the style condition; the mean content-minus-style difference is −.467 (exact two-sided $p = .03125$). No reliable paired condition difference appears for the lexical or hybrid methods. This initial pattern is consistent with the prompts, but it does not establish that the generator transferred an abstract style independently of wording.

4.2 What Survives Phrase Masking?

Table 2 answers RQ3. Content recovery is highly robust: strict 3-gram masking reduces Word TF–IDF MRR only from .967 to .944. Content adaptations therefore retain distributed semantic evidence beyond copied phrases. Style recovery is more sensitive, yet Word TF–IDF remains at MRR .598 after 3-gram masking, well above the .024 random expectation. Remaining unigrams, bigrams, and dispersed lexical choices still reveal the source even after longer shared sequences are neutralized. The batch-level Word TF–IDF decline is descriptive rather than reliable with six pairs ($p = .125$).

Stylometry behaves differently. Masking only shared 5-grams reduces style MRR from .540 to .049; strict 3-gram masking reduces it to .018. Both paired drops are exact $p = .03125$. Thus, the original stylometric result is largely carried by function and frequent words embedded in source-local phrase frames. The structural-only profile retains MRR .267 after strict masking, but its median rank is 78.5 and its success is concentrated in two Gemini–source cells. We therefore find limited, heterogeneous structural evidence rather than a general abstract style signature.

System-level diagnostics reinforce this interpretation. Style passages share 2.3% of their distinct 5-grams with the source for ChatGPT and 25.9% for Gemini. Under strict 3-gram masking, style Word TF–IDF MRR is .318 for ChatGPT and .878 for Gemini. These are descriptive differences because there is only one batch per system–source–condition cell. Nevertheless, they show that the aggregate result is not a stable property of all generators: one interface frequently maps the storm scenario into source-local lexical and rhetorical frames despite instructions not to reuse recognizable phrases.

5 Discussion and Limitations

The practical finding is encouraging for digital-library provenance support. A generated derivative can serve as a query that returns candidate source records without model access, supervised labels, or generation-time logs. In a collection interface, these rankings should be presented with evidence such as shared phrases and treated as curator-facing link suggestions. They are not proof of authorship, plagiarism, or training-data membership.

The ablation also changes how the style result should be interpreted. Topic replacement does not ensure source independence: a passage can discuss an unrelated storm while retaining local phrase templates, function-word sequences, and isolated lexical choices. High recovery can therefore indicate both useful provenance evidence and incomplete compliance with a "style-only" instruction. Systems that preserve generation metadata should distinguish semantic derivation from stylistic conditioning rather than collapsing both into a single provenance relation.

This study is exploratory and has important limitations. It evaluates only two displayed system configurations, three source chapters, and one homogeneous English-language collection. The five passages returned within each response may also be correlated. The results therefore demonstrate source recovery in a controlled, closed collection but do not establish scalability or transferability to larger, heterogeneous, multilingual, or open-world digital libraries.

The study also focuses on transparent lexical and stylometric methods and does not compare them with modern dense or neural retrievers. Future work should expand the benchmark to non-canonical collections, independent generation calls, stronger transformations, multilingual material, and derivatives based on multiple sources. Comparisons with semantic retrieval models will also be necessary to determine when lexical residual traces remain useful beyond this controlled setting.

6 Reproducibility and AI Use

Generative AI systems produced the experimental passages and assisted with code and manuscript preparation. The author inspected the data, executed and checked the analyses, and remains responsible for all claims.

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